

Numerical Methods

BCA — Semester IV — 2025 — Answer Sheet
Subject Code: CACS252

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

2. What do you mean by true error? Find a positive root of equation $(x^2 = 1)$, which lies between 0 and 1 using bisection method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is interpolation? Given the following set of data points, obtained the table of divided difference. Use the table to estimate the value of $f(1.5)$.

x_i	0	1	2	3	4
$f(x_i)$	0.1234	0.2345	0.3456	0.4567	0.5678

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Solve the following system of equations by Gauss elimination method.

$$\begin{cases} 2x + y + 4z = 12 \\ 8x - 3y + 2z = 23 \\ 4x + 11y - z = 33 \end{cases}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. The distance of motorbike at interval 2 minutes are given below:

Time(minutes)	Distance(Km)
0	0.1
2	0.53
4	0.86
6	0.7

Evaluate the velocity and acceleration of motorbike at time 2.37 and 4.57 minutes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Write a program to compute value of y when $x = 2$ such that $\frac{dy}{dx} = 3x^2 + 1$ with $y(1) = 2$ and $h = 0.25$ using Euler's method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Find the largest Eigen value and corresponding Eigen vector of the following matrix using power method.

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the Poisson equation $\nabla^2 f = 2x^2y^2$ over the square domain $(0 \leq x \leq 3)$ and $(0 \leq y \leq 3)$ with $(f = 0)$ on the boundary and $(h = 1)$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Group C

Attempt any TWO questions[2x10=20]

9. What do you mean by ill condition? Solve the following system using Jacobi iteration method.
 $(2a + b + c = 5)$
 $(3a + 5b + 2c = 15)$
 $(2a + b + 4c = 8)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. How interpolation is differ from regression? Write down algorithm and program for Lagrange interpolation polynomial.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is fixed point iteration method? How it converge the root of non-linear equation? Find the root of equation $(f(x) = x^2 - 3x + 2)$ using N-R method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.